PYTHON DAY 3

Comparisons, Boolean Logic, and Decision Making

Total time: 3 to 4 hours | **Main goal:** Make your Python programs choose different paths.

Today's Big Idea

Programs can now evaluate conditions and choose different paths. Today you learn comparison operators, booleans, truthiness, if/elif/else, and/or/not, testing boundaries, and debugging conditional logic.

```
Input / Data
↓
Comparison
↓
True or False
↓
if / elif / else
↓
Different program behavior
```

1. Warm-up From Memory - 15 minutes

Create warmup.py without opening Day 2. Ask for name and age, convert age to int, print age after 5 years, and print type(age).

2. Comparison Operators - 35 minutes

Create comparisons.py and predict every result before running.

```
a = 10
b = 20

print(a == b)
print(a != b)
print(a < b)
print(a > b)
print(a <= b)
print(a >= b)
```

Important: = means assignment. == means comparison.

```
print("apple" == "apple")
print("apple" == "Apple")
print("10" == 10)
```

Research task: investigate the high-level difference between == and is using the official Python documentation.

Official reading: https://docs.python.org/3/library/stdtypes.html#comparisons

3. Boolean Values and Truthiness - 30 minutes

```
print(bool(True))
print(bool(False))
print(bool(0))
print(bool(1))
print(bool(''))
print(bool('Python'))
print(bool(None))
```

Before looking up the full explanation, write your prediction and observations. Research: What is truthiness? How is False different from a value that evaluates as false?

4. Your First if Statement - 35 minutes

```
age = 20

if age >= 18:
print('You are an adult')
```

Change age to values above, equal to, and below 18. Notice that indentation is part of Python syntax. Deliberately break the indentation and read the error.

```
age = 20
if age >= 18:
print('You are an adult')
```

5. if / else - 25 minutes

```
age = int(input('Enter your age: '))

if age >= 18:
print('Adult')
else:
print('Minor')
```

Test 17, 18, and 19. Explain why 18 belongs to the first branch.

6. if / elif / else - 30 minutes

```
score = int(input('Enter score: '))

if score >= 90:
print('Excellent')
elif score >= 75:
print('Good')
elif score >= 50:
print('Pass')
else:
print('Needs improvement')
```

Boundary test: 49, 50, 74, 75, 89, 90. Research why the order of conditions matters and deliberately place them in a bad order to observe the behavior.

7. Boolean Logic: and, or, not - 40 minutes

```
age = 25
has_ticket = True

print(age >= 18 and has_ticket)
print(age < 18 or has_ticket)
print(not has_ticket)
```

```
print(True and True)
print(True and False)
print(False and True)
print(False and False)

print(True or True)
print(True or False)
print(False or True)
print(False or False)
```

Build your own truth table. Then research short-circuit behavior: if the first part of an AND expression is already false, why might Python not need to evaluate the second part?

8. Mini Project - Smart Eligibility Checker

Create eligibility.py. Ask for age, whether the user has a valid document, and whether they meet a minimum score. Decide eligibility. Use at least one comparison, one and, one or, and if/elif/else. Test at least 8 combinations.

9. Practical Project - CLI Login Simulator

Build login_simulator.py. Hard-code a username and password for now. Ask the user for both and handle multiple outcomes.

```
correct_username = "admin"
correct_password = "python123"

username = input("Username: ")
password = input("Password: ")

if username == correct_username and password == correct_password:
print("Login successful")
else:
print("Invalid credentials")
```

Improve it to distinguish: unknown username, correct username with wrong password, successful login, and empty input. Before coding, list possible input paths as test cases.

10. Boundary Testing Lab - 25 minutes

Create a program that classifies a number as negative, zero, or positive. Test -1, 0, and 1. Then design your own age categories and test every boundary.

Record: Input | Expected output | Actual output | Pass/Fail.

11. Debugging Exercises - 20 minutes

```
age = 20
if age = 20:
print('Twenty')
```

```
age = 20
if age > 18
print('Adult')
```

```
is_member = 'False'
if is_member:
print('Member')
```

For each error or surprise, identify the exception or behavior, the exact cause, and the fix. Investigate why the string 'False' is truthy.

12. Main Project - Intelligent Discount Calculator

Build discount_calculator.py. Ask for product price, customer type, and whether the customer has a promotional code. Define clear business rules for discounts and implement them.

• Use float() for price.

• Use at least three branches.

• Use and/or meaningfully.

• Handle invalid customer types.

• Test price 0 and exact discount boundaries.

• Print original price, discount amount, and final price.

• Write test cases before finalizing.

13. Knowledge Check

Q1. What is the difference between = and ==?

Q2. What is the difference between == and is at a high level?

Q3. What type does a comparison expression produce?

Q4. Why does elif order matter?

Q5. What is truthiness?

Q6. What is False and True?

Q7. What is False or True?

Q8. What does not do?

Q9. Why is indentation important?

Q10. What is boundary testing?

Q11. Why is the string 'False' truthy?

14. Day 3 Completion Checklist

- [ ] I understand = versus ==.

- [ ] I can use ==, !=, <, >, <=, >=.

- [ ] I understand True and False.

- [ ] I investigated truthiness.

- [ ] I can write if statements.

- [ ] I can use if/elif/else.

- [ ] I can use and, or, not.

- [ ] I fixed conditional syntax/indentation errors.

- [ ] I completed Eligibility Checker.

- [ ] I completed CLI Login Simulator.

- [ ] I completed Intelligent Discount Calculator.

- [ ] I recorded boundary test cases.

Suggested Study Schedule

15 min warm-up | 35 min comparisons | 30 min booleans | 35 min if | 25 min else | 30 min elif/boundaries | 40 min boolean logic | 30 min mini project | 45 min login project | 25 min testing | 20 min debugging | 45 min main project.

Official References

Python Tutorial: https://docs.python.org/3/tutorial/
Comparisons: https://docs.python.org/3/library/stdtypes.html#comparisons
Boolean operations: https://docs.python.org/3/library/stdtypes.html#boolean-operations-and-or-not
Built-in types: https://docs.python.org/3/library/stdtypes.html

Day 4 Preview

Next: while loops, for loops, range(), iteration, break, continue, nested loops, and common infinite-loop mistakes. Your programs will start repeating useful work without copy-pasting code.